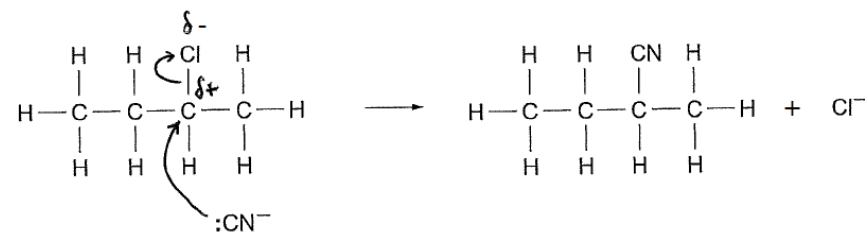


Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
10.	(a)	(i)		 δ^- δ^+ :CN^- $\text{H}-\text{C}-\text{C}-\text{C}-\text{C}-\text{H} \longrightarrow \text{H}-\text{C}-\text{C}-\text{C}-\text{C}-\text{H} + \text{Cl}^-$ H H CN H $$ $$ $$ $$ H H H H δ^+ and δ^- shown on C and Cl (1) arrow from lone pair on CN^- to δ^+ C (1) arrow from C—Cl bond to Cl (1) 2-cyanobutane and Cl^- as products (1)	1					
					1	1		4		
		(ii)		nucleophilic substitution	1			1		